💊 MediRx Mobile App

Complete RESTful API Documentation

API Reference · Patient & Doctor Applications · 15 Modules

Document detail	Value
:--	:--
Application	`medirx_patient`
API style	REST
Authentication	Bearer token (JWT)
Payload formats	`application/json` · `multipart/form-data`
Coverage	Authentication, patient care, doctor tools, scheduling, records, and settings

***Implementation note:** URLs, credentials, identifiers, and response payloads in this document are examples from the specification. Confirm the actual backend base URL, endpoint availability, and response schema before connecting a production app.*

This document provides a comprehensive RESTful API specification tailored to the **MediRx** mobile application (`medirx_patient`). It covers both **Patient** and **Doctor** roles across all 15 system modules, complete with HTTP methods, request headers, payload parameters, query options, status codes, and realistic dummy JSON responses.

1. Global API Architecture & Standards

1.1 Base URL & Versioning

- **Production URL**: `https://medirx.unipexel.org/public/app`
- **Staging / Mock URL**: `https://sandbox.medirx.unipexel.org/public/app`
- **API Versioning**: Prefix `/api/v1`

1.2 Authentication & Security

- **Type**: Bearer Token (JWT - JSON Web Token)
- **Header**: `Authorization: Bearer <jwt_access_token>`
- **Content Types**:
- `application/json` for standard payload exchange
- `multipart/form-data` for file, document, and audio uploads

1.3 Standard Response Format

All responses adhere to a consistent JSON envelope:

Standard Success Response

```
{
  "success": true,
  "statusCode": 200,
  "message": "Operation successful",
  "data": {}
}
```

Standard Paginated Response

```
{
  "success": true,
  "statusCode": 200,
  "message": "Items fetched successfully",
  "data": [],
  "pagination": {
    "totalItems": 150,
    "itemCount": 10,
    "itemsPerPage": 10,
    "totalPages": 15,
    "currentPage": 1
  }
}
```

Standard Error Response

```
{
  "success": false,
  "statusCode": 400,
  "error": "BAD_REQUEST",
  "message": "Validation failed: Phone number is invalid"
}
```

---

2. Module Index

1. [Auth & Identity Module (`/auth`)](#module-1-auth--identity-module)
2. [Patient Dashboard & Vitals (`/dashboard`)](#module-2-patient-dashboard--vitals)
3. [Medication & Prescription (`/medication`)](#module-3-medication--prescription)
4. [Doctor Portal & Clinical Management (`/doctor`)](#module-4-doctor-portal--clinical-management)
5. [Scheduling & Appointments (`/scheduling`)](#module-5-scheduling--appointments)
6. [Vaccine Tracker (`/vaccines`)](#module-6-vaccine-tracker)
7. [Pediatric Dose Calculator (`/pediatric-dose`)](#module-7-pediatric-dose-calculator)
8. [Pregnancy Tracking (`/pregnancy`)](#module-8-pregnancy-tracking)
9. [Menstrual Cycle & Fertility (`/menstrual-cycle`)](#module-9-menstrual-cycle--fertility)
10. [Nearest Medical Facilities (`/nearest-medical`)](#module-10-nearest-medical-facilities)
11. [Doctor Ratings & Reviews (`/doctor-ratings`)](#module-11-doctor-ratings--reviews)
12. [AI Doctor Assistant (`/ai-assistant`)](#module-12-ai-doctor-assistant)
13. [Notification Center (`/notifications`)](#module-13-notification-center)
14. [Patient Profile & Health Records (`/profile`)](#module-14-patient-profile--health-records)
15. [Settings & App Preferences (`/settings`)](#module-15-settings--app-preferences)

🔐 Module 1: Auth & Identity Module

Handles user onboarding, multi-role registration (Doctor & Patient), credential authentication, biometric sign-in, and password lifecycle.

1.1 User Login

- **Endpoint**: `POST /auth/login`
- **Description**: Authenticate via Health ID / Phone number and password.

Request Body

```
{
  "identifier": "+8801712345678",
  "password": "Password@123"
}
```

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Login successful",
  "data": {
      "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6InBhdC0wMDEiLCJyb2xlIjoicGF0aWVudCJ9...",
      "refreshToken": "d7a8e29f-3b7c-48c9-a9a3-32454a7c00e1"
  }
}
```

1.3 Patient Registration

- **Endpoint**: `POST /auth/register/patient`
- **Description**: Registers a new patient.

Request Body

```
{
  "fullName": "Rahim Ahmed",
  "phone": "+8801712345678",
  "password": "SecurePassword123!"
}
```

Response `201 Created`

```
{
  "success": true,
  "statusCode": 201,
  "message": "Registration successful. Welcome to MediRx!",
  "data": {
      "accessToken": "eyJhbGciOiJIUzI1NiIs...",
      "refreshToken": "fb89a421-2e6b-4f0e-b83c-15a0129bc8de"
  }
}
```

1.4 Doctor Registration

- **Endpoint**: `POST /auth/register/doctor`
- **Description**: Registers a medical doctor with BMDC validation.

Request Body

```
{
  "fullName": "Dr. Arif Rahman",
  "bmdcNumber": "BMDC-12345",
  "email": "doctor@medirx.com",
  "phone": "+8801711234567",
  "password": "SecureDoctorPass123!"
}
```

Response `201 Created`

```
{
  "success": true,
  "statusCode": 201,
  "message": "Registration successful. Welcome to MediRx!",
  "data": {
      "accessToken": "eyJhbGciOiJIUzI1NiIs...",
      "refreshToken": "fb89a421-2e6b-4f0e-b83c-15a0129bc8de"
  }
}
```

1.5 Password Recovery & Reset

- **Endpoint**: `POST /auth/forgot-password`

Request Body

```
{
  "phoneOrEmail": "+8801712345678"
}
```

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Verification code dispatched to your registered phone number",
  "data": {
    "otpSessionId": "otp_sess_891238472918",
    "expiresInSeconds": 180
  }
}
```

1.6 Fetch Authenticated User Profile Using Access Token

Use the access token returned by login to request the signed-in user's profile. The profile route below reuses the documented personal-information endpoint.

Endpoint: GET /profile/personal-info

Headers

```
Authorization: Bearer <access_token>
Accept: application/json
```

Example Response 200 OK

```
{
  "success": true,
  "message": "Profile fetched successfully",
  "data": {
    "id": "user_id",
    "name": "Example User",
    "email": "user@example.com",
    "phone": "+8801XXXXXXXXX"
  }
}
```

Note: The response fields above are illustrative. Match the exact field names and response envelope returned by the MediRx backend.

1.7 Get a New Access Token Using a Refresh Token

When the access token expires, send the refresh token to the backend refresh endpoint to request a new access token. Keep refresh tokens secure and do not log them.

Proposed endpoint (confirm with backend): POST /auth/refresh

Request Headers

```
Content-Type: application/json
Accept: application/json
```

Request Body

```
{
  "refresh_token": "<refresh_token>"
}
```

Example Response 200 OK

```
{
  "access_token": "<new_access_token>",
  "refresh_token": "<new_refresh_token>"
}
```

Implementation notes

- • The /auth/refresh route, request key, response fields, and token lifetime are examples and must be verified against the actual backend.
- • If the backend rotates refresh tokens, securely replace the stored refresh token with the newly returned value.
- • If refresh fails (for example, the refresh token is expired or revoked), clear local credentials and ask the user to sign in again.

📊 Module 2: Patient Dashboard & Vitals

Feeds the primary patient home screen, including multi-vital trend charts, medication schedule, monthly consistency heatmap, and quick module links.

2.1 Get Patient Dashboard Overview

- **Endpoint**: `GET /dashboard/overview`
- **Headers**: `Authorization: Bearer <token>`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Dashboard data retrieved",
  "data": {
    "patient": {
      "id": "pat-001",
      "name": "Druvo",
      "healthId": "RX-987654",
      "greeting": "Good morning, Druvo"
    },
    "adherenceRate": 85.0,
    "todayMedsSummary": {
      "total": 3,
      "taken": 1,
      "pending": 2
    },
    "todayMedications": [
      {
        "id": "1",
        "name": "Atorvastatin",
        "dosage": "10mg",
        "time": "08:00 AM",
        "isTaken": true,
        "instructions": "Take before bed"
      },
      {
        "id": "2",
        "name": "Metformin",
        "dosage": "500mg",
        "time": "01:00 PM",
        "isTaken": false,
        "instructions": "Take after lunch"
      },
      {
        "id": "3",
        "name": "Vitamin D3",
        "dosage": "2000 IU",
        "time": "08:00 PM",
        "isTaken": false,
        "instructions": "Take with dinner"
      }
    ],
    "quickStats": {
      "activePrescriptions": 2,
      "upcomingAppointments": 1,
      "pendingVaccinations": 1
    }
  }
}
```

2.2 Get 7-Day Vitals Trend

- **Endpoint**: `GET /dashboard/vitals-trend`
- **Query Parameters**: `days=7`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Vitals trends retrieved successfully",
  "data": {
    "dates": ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"],
    "metrics": {
      "heartRate": {
        "unit": "bpm",
        "colorHex": "#F44336",
        "values": [72, 75, 71, 78, 74, 72, 76]
      },
      "bloodPressureSystolic": {
        "unit": "mmHg",
        "colorHex": "#2196F3",
        "values": [120, 122, 118, 125, 121, 119, 120]
      },
      "bloodPressureDiastolic": {
        "unit": "mmHg",
        "colorHex": "#64B5F6",
        "values": [80, 82, 78, 84, 80, 79, 81]
      },
      "spo2": {
        "unit": "%",
        "colorHex": "#00BCD4",
        "values": [98, 97, 99, 98, 98, 99, 98]
      },
      "glucose": {
        "unit": "mg/dL",
        "colorHex": "#9C27B0",
        "values": [105, 110, 98, 115, 102, 108, 104]
      },
      "temperature": {
        "unit": "°F",
        "colorHex": "#FF9800",
        "values": [98.4, 98.6, 98.2, 98.5, 98.7, 98.4, 98.6]
      }
    }
  }
}
```


2.3 Get Medication Consistency Heatmap

- **Endpoint**: `GET /dashboard/heatmap?month=2&year=2026`

Response `200 OK`


```
{
  "success": true,
```

```
  "statusCode": 200,
  "message": "Monthly heatmap retrieved",
  "data": {
    "month": 2,
    "year": 2026,
    "monthName": "February 2026",
    "totalDays": 28,
    "heatmap": [
      { "day": 1, "date": "2026-02-01", "consistency": 1.0, "status": "all_taken" },
      { "day": 2, "date": "2026-02-02", "consistency": 1.0, "status": "all_taken" },
      { "day": 3, "date": "2026-02-03", "consistency": 1.0, "status": "all_taken" },
      { "day": 4, "date": "2026-02-04", "consistency": 1.0, "status": "all_taken" },
      { "day": 5, "date": "2026-02-05", "consistency": 1.0, "status": "all_taken" },
      { "day": 6, "date": "2026-02-06", "consistency": 0.4, "status": "partially_taken" },
      { "day": 7, "date": "2026-02-07", "consistency": 0.4, "status": "partially_taken" },
      { "day": 14, "date": "2026-02-14", "consistency": 0.8, "status": "partially_taken" },
      { "day": 20, "date": "2026-02-20", "consistency": 0.1, "status": "missed" }
    ]
  }
}
```

2.4 Get Date-Specific Medication Breakdown

- **Endpoint**: `GET /dashboard/day-details?date=2026-02-14`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Day log details retrieved",
  "data": {
    "date": "2026-02-14",
    "formattedDate": "February 14, 2026",
    "items": [
      {
        "name": "Atorvastatin",
        "dosage": "10mg",
        "time": "08:00 AM",
        "isTaken": true,
        "status": "Taken"
      },
      {
        "name": "Metformin",
        "dosage": "500mg",
        "time": "01:00 PM",
        "isTaken": true,
        "status": "Taken"
      },
      {
        "name": "Vitamin D3",
        "dosage": "2000 IU",
        "time": "08:00 PM",
        "isTaken": false,
        "status": "Missed"
      }
    ]
  }
}
```

💊 Module 3: Medication & Prescription

Covers active medications, pill intake toggling, refills, manual prescription creation, e-prescription download, and AI/OCR scan analysis.

3.1 Get Current Medications

- **Endpoint**: `GET /medications/active`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Current medications fetched",
  "data": {
    "currentMedications": [
      {
        "id": "med-01",
        "name": "Atorvastatin",
        "dosage": "10mg",
        "frequency": "Once daily",
        "instructions": "Before Bed",
        "icon": "pill",
        "accentColor": "#2196F3",
        "remainingDays": 22,
        "isRefillNeeded": false
      },
      {
        "id": "med-02",
        "name": "Metformin",
        "dosage": "500mg",
        "frequency": "Twice daily",
        "instructions": "After Meal",
        "icon": "tablet",
        "accentColor": "#4CAF50",
        "remainingDays": 14,
        "isRefillNeeded": false
      }
    ],
    "refillsNeeded": [
      {
        "id": "med-03",
        "name": "Vitamin D3",
        "dosage": "2000 IU",
        "frequency": "Once daily",
        "instructions": "Take with meal",
        "daysLeft": 2,
        "icon": "pill",
        "accentColor": "#FF9800",
        "isWarning": true
      }
    ]
  }
}
```

3.2 Toggle Medication Status

- **Endpoint**: `PATCH /medications/tasks/:id/toggle`

Request Body

```
{
  "date": "2026-10-13",
  "isTaken": true
}
```

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Medication marked as taken",
  "data": {
    "taskId": "1",
    "name": "Atorvastatin",
    "isTaken": true,
    "timestamp": "2026-10-13T08:05:22Z"
  }
}
```

3.3 Get Prescription History

- **Endpoint**: `GET /medications/prescriptions`

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Prescriptions fetched",
  "data": [
    {
      "id": "rx-101",
      "doctor": "Dr. Sarah Ahmed",
      "specialization": "Cardiologist",
      "date": "05 Feb 2026",
      "diagnosis": "Hypertension",
      "medicationCount": 3,
      "pdfUrl": "https://api.medirx.com/prescriptions/rx-101.pdf"
    },
    {
      "id": "rx-102",
      "doctor": "Dr. John Smith",
      "specialization": "Endocrinologist",
      "date": "28 Jan 2026",
      "diagnosis": "Type 2 Diabetes",
      "medicationCount": 2,
      "pdfUrl": "https://api.medirx.com/prescriptions/rx-102.pdf"
    }
  ]
}
```

3.4 Create Prescription Manually

- **Endpoint**: `POST /medications/prescriptions/manual`

Request Body

```
{
  "patient_uid": “550e8400-e29b-41d4-a716-446655440000”,
  "doctor_uuid":”550e8400-e29b-41d4-a716-446655440000”
  "prescriptionDate": "2026-10-13",
  "diagnosis": "Mild Seasonal Bronchitis",
  "advice": "Drink lukewarm water, avoid cold beverages",
  "followUpDate": "2026-10-27",
  "medications": [
    {
      "name": "Amoxicillin",
      "dosage": "500mg",
      "form": "Capsule",
      "frequency": "Three times daily (1-1-1)",
      "duration": "7 Days",
      "instructions": "After meal"
    },
    {
      "name": "Cetirizine",
      "dosage": "10mg",
      "form": "Tablet",
      "frequency": "Once daily at night (0-0-1)",
      "duration": "5 Days",
      "instructions": "Before sleep"
    }
  ]
}
```

Response `201 Created`

```
{
  "success": true,
  "statusCode": 201,
  "message": "Prescription created and saved successfully",
  "data": {
    "prescriptionId": "rx-manual-90214",
    "ePrescriptionCode": "RX-MAN-90214",
    "downloadUrl": "https://api.medirx.com/prescriptions/rx-manual-90214.pdf"
  }
}
```

3.5 Fetch E-Prescription by Code

- **Endpoint**: `GET /medications/e-prescription/:code`
- **Example Route**: `/medications/e-prescription/RX-12345678`

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Prescription retrieved successfully",
  "data": {
    "id": "rx-001",
```

```
    "code": "RX-12345678",
    "doctor": {
      "id": "doc-001",
      "name": "Dr. Arif Rahman",
      "qualification": "MBBS, FCPS (Medicine)",
      "hospital": "Dhaka Medical College & Hospital"
    },
    "patient": {
      "name": "Rahim Ahmed",
      "age": 45,
      "gender": "Male"
    },
    "date": "2026-09-10",
    "diagnosis": "Hypertension & Type 2 Diabetes",
    "items": [
      {
        "medicineName": "Metformin",
        "dosage": "500mg",
        "frequency": "Twice daily",
        "duration": "1 month",
        "instructions": "Take after meals"
      },
      {
        "medicineName": "Amlodipine",
        "dosage": "5mg",
        "frequency": "Once daily",
        "duration": "1 month",
        "instructions": "Take in the morning"
      }
    ],
    "notes": "Monitor blood pressure daily. Reduce salt intake."
  }
}
```

3.6 OCR / AI Prescription Image Scanner

- **Endpoint**: `POST /medications/prescriptions/scan-ocr`
- **Content-Type**: `multipart/form-data`

Request Payload

- `file`: `prescription_photo.jpg`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Prescription scanned and parsed by AI model",
  "data": {
    "extractedDoctor": "Dr. K. M. Hossain",
    "extractedDate": "2026-10-10",
    "confidenceScore": 0.94,
    "parsedMedicines": [
      {
        "name": "Pantoprazole",
        "genericName": "Pantoprazole Sodium",
        "strength": "40mg",
        "form": "Tablet",
        "dosage": "1 Tab",
```

```
        "frequency": "Once daily",
        "timing": "Before breakfast",
        "duration": "14 Days"
      },
      {
        "name": "Paracetamol",
        "genericName": "Acetaminophen",
        "strength": "500mg",
        "form": "Tablet",
        "dosage": "1 Tab",
        "frequency": "SOS / As needed for fever",
        "timing": "After meals",
        "duration": "3 Days"
      }
    ],
    "rawText": "Rx:\n1. Tab. Pantoprazole 40mg 1+0+0 before meal 14 days\n2. Tab. Paracetamol 500mg as needed
for fever"
  }
}
```

🩺 Module 4: Doctor Portal & Clinical Management

Provides doctor dashboard, patient lists, complete patient profile with 5 sub-tabs (Overview, Medications, Prescriptions, Adherence, Clinical Notes), prescription builder, and analytics.

4.1 Doctor Profile Details

- **Endpoint**: `GET /doctor/profile`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Doctor profile retrieved",
  "data": {
    "id": "doc-001",
    "name": "Dr. Arif Rahman",
    "qualification": "MBBS, FCPS (Medicine)",
    "specialization": "Internal Medicine",
    "regNo": "BMDC-12345",
    "hospital": "Dhaka Medical College & Hospital",
    "experience": "12 years",
    "rating": 4.8,
    "totalPatients": 320,
    "phone": "+880 1711-234567",
    "email": "doctor@medirx.com",
    "bio": "Dr. Arif Rahman is a highly experienced Internal Medicine specialist with over 12 years of
clinical practice."
  }
}
```

4.2 Doctor Dashboard Home

- **Endpoint**: `GET /doctor/dashboard`

Response 200 OK


```
{
  "success": true,
  "statusCode": 200,
  "message": "Doctor dashboard data loaded",
  "data": {
    "stats": {
      "todayPatients": 8,
      "totalPrescriptions": 1420,
      "activePatients": 320,
      "avgAdherence": 76.5
    },
    "todaySchedule": [
      {
        "id": "sch-01",
        "patientId": "pat-001",
        "patientName": "Rahim Ahmed",
        "time": "10:00 AM",
        "type": "Follow-up",
        "status": "upcoming"
      },
      {
        "id": "sch-02",
        "patientId": "pat-002",
        "patientName": "Fatima Begum",
        "time": "11:30 AM",
        "type": "Consultation",
        "status": "upcoming"
      }
    ],
    "recentPrescriptions": [
      {
        "id": "rx-001",
        "patientId": "pat-001",
        "patientName": "Rahim Ahmed",
        "date": "2026-09-10",
        "itemCount": 3,
        "status": "active"
      }
    ]
  }
}
```

4.3 Patient Directory (Search & Filter)

- **Endpoint**: `GET /doctor/patients`
- **Query Parameters**: `search=Rahim&condition=Diabetes&page=1&limit=20`

Response 200 OK


```
{
  "success": true,
  "statusCode": 200,
  "message": "Patients list retrieved",
  "data": [
    {
      "id": "pat-001",
      "name": "Rahim Ahmed",
      "age": 45,
```

```
      "gender": "Male",
      "bloodGroup": "B+",
      "conditions": ["Hypertension", "Diabetes Type 2"],
      "phone": "+880 1712-345678",
      "address": "Mirpur-10, Dhaka",
      "lastVisit": "2026-09-10T00:00:00Z",
      "adherencePercent": 78.0,
      "avatarInitials": "RA"
    },
    {
      "id": "pat-002",
      "name": "Fatima Begum",
      "age": 32,
      "gender": "Female",
      "bloodGroup": "A+",
      "conditions": ["Anemia", "Hypothyroidism"],
      "phone": "+880 1713-456789",
      "address": "Dhanmondi, Dhaka",
      "lastVisit": "2026-09-15T00:00:00Z",
      "adherencePercent": 92.0,
      "avatarInitials": "FB"
    }
  ],
  "pagination": {
    "total": 320,
    "page": 1,
    "pageSize": 20
  }
}
```

4.4 Patient 360 View (Patient Detail Tabs)

- **Endpoint**: `GET /doctor/patients/:patientId/details`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Patient detailed profile loaded",
  "data": {
    "patient": {
      "id": "pat-001",
      "name": "Rahim Ahmed",
      "age": 45,
      "gender": "Male",
      "bloodGroup": "B+",
      "conditions": ["Hypertension", "Diabetes Type 2"],
      "adherencePercent": 78.0
    },
    "activeMedications": [
      {
        "medicineName": "Metformin",
        "dosage": "500mg",
        "frequency": "Twice daily",
        "prescribedDate": "2026-09-10"
      },
      {
        "medicineName": "Amlodipine",
        "dosage": "5mg",
        "frequency": "Once daily",
```

```
        "prescribedDate": "2026-09-10"
      }
    ],
    "prescriptions": [
      {
        "id": "rx-001",
        "date": "2026-09-10",
        "status": "active",
        "notes": "Monitor blood pressure daily. Reduce salt intake."
      }
    ],
    "adherenceCalendar": [
      { "date": "2026-10-01", "status": "taken" },
      { "date": "2026-10-02", "status": "missed" },
      { "date": "2026-10-03", "status": "taken" }
    ],
    "clinicalNotes": [
      {
        "id": "note-01",
        "date": "2026-09-10",
        "title": "Routine Consultation",
        "content": "Patient reported slight morning fatigue. Fasting blood glucose is 125 mg/dL. Renal
profile normal.",
        "isPrivate": false
      }
    ]
  }
}
```


4.5 Medicine Search & Autocomplete

- **Endpoint**: `GET /doctor/medicines/search?query=met`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Medicine suggestions retrieved",
  "data": [
    {
      "id": "m-01",
      "name": "Metformin",
      "genericName": "Metformin HCl",
      "strength": "500mg",
      "form": "Tablet",
      "manufacturer": "Square"
    },
    {
      "id": "m-20",
      "name": "Metronidazole",
      "genericName": "Metronidazole",
      "strength": "400mg",
      "form": "Tablet",
      "manufacturer": "Globe"
    }
  ]
}
```

4.6 Issue Digital Prescription (Doctor)

- **Endpoint**: `POST /doctor/prescriptions`
- **Content-Type**: `multipart/form-data` or `application/json`

Request Body

```
{
  "patientId": "pat-001",
  "items": [
    {
      "medicineId": "m-01",
      "medicineName": "Metformin",
      "dosage": "500mg",
      "frequency": "Twice daily",
      "duration": "1 month",
      "instructions": "Take after meals"
    },
    {
      "medicineId": "m-03",
      "medicineName": "Amlodipine",
      "dosage": "5mg",
      "frequency": "Once daily",
      "duration": "1 month",
      "instructions": "Take in the morning"
    }
  ],
  "notes": "Reduce daily carbohydrate intake. Walk 30 minutes every morning.",
  "voiceInstructions": "Take Metformin with meals twice a day. Never skip Amlodipine.",
  "status": "active"
}
```

Response `201 Created`

```
{
  "success": true,
  "statusCode": 201,
  "message": "Prescription issued successfully",
  "data": {
    "prescriptionId": "rx-2026-09104",
    "pdfDownloadUrl": "https://api.medirx.com/prescriptions/rx-2026-09104.pdf",
    "qrCode": "https://api.medirx.com/qr/rx-2026-09104.png"
  }
}
```

4.7 Doctor Clinical Analytics & Metrics

- **Endpoint**: `GET /doctor/analytics`

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Doctor analytics retrieved",
  "data": {
    "totalPrescriptions": 1420,
    "activePatients": 320,
    "thisMonthPrescriptions": 84,
```

```
    "avgAdherence": 76.5,
    "topMedicines": [
      { "medicineName": "Metformin", "prescriptionCount": 245, "colorHex": "#009688" },
      { "medicineName": "Atorvastatin", "prescriptionCount": 198, "colorHex": "#26A69A" },
      { "medicineName": "Omeprazole", "prescriptionCount": 175, "colorHex": "#80CBC4" },
      { "medicineName": "Amlodipine", "prescriptionCount": 142, "colorHex": "#B2DFDB" }
    ],
    "prescriptionsByMonth": {
      "May": 65,
      "Jun": 72,
      "Jul": 80,
      "Aug": 95,
      "Sep": 88,
      "Oct": 84
    }
  }
}
```

📅 Module 5: Scheduling & Appointments

Powers patient daily schedule timelines and doctor appointment reservations.

5.1 Get Daily Schedule Timeline

- **Endpoint**: `GET /scheduling/daily-timeline?date=2026-10-13`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Daily schedule loaded",
  "data": {
    "selectedDate": "2026-10-13",
    "timeline": [
      {
        "id": "sch-01",
        "title": "Morning Medicine",
        "time": "08:00 AM",
        "subtitle": "Atorvastatin (10mg)",
        "icon": "sun",
        "category": "medication",
        "colorHex": "#FF9800",
        "status": "completed"
      },
      {
        "id": "sch-02",
        "title": "Doctor Appointment",
        "time": "11:30 AM",
        "subtitle": "Dr. Sarah Johnson (Cardiologist)",
        "icon": "calendar",
        "category": "appointment",
        "colorHex": "#2196F3",
        "status": "upcoming"
      },
      {
        "id": "sch-03",
        "title": "Blood Pressure Check",
        "time": "06:00 PM",
        "subtitle": "Daily tracking",
```

```
        "icon": "activity",
        "category": "vital_check",
        "colorHex": "#F44336",
        "status": "pending"
      }
    ]
  }
}
```

5.2 Book Doctor Appointment

- **Endpoint**: `POST /scheduling/appointments`

Request Body

```
{
  "doctorId": "doc-001",
  "date": "2026-10-20",
  "timeSlot": "11:30 AM",
  "reason": "Routine hypertension checkup",
  "type": "in_person"
}
```

Response `201 Created`

```
{
  "success": true,
  "statusCode": 201,
  "message": "Appointment booked successfully",
  "data": {
    "appointmentId": "apt-89412",
    "doctorName": "Dr. Sarah Johnson",
    "dateTime": "2026-10-20T11:30:00Z",
    "chamber": "Suite 402, City General Hospital",
    "status": "confirmed"
  }
}
```

💉 Module 6: Vaccine Tracker

Tracks completed vaccinations, booster shots, countdowns, and immunization records.

6.1 Get Vaccine Schedule

- **Endpoint**: `GET /vaccines/schedule`

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Vaccine schedule retrieved",
  "data": {
    "nextVaccinationCountdown": "Next vaccination in 12 days",
    "upcomingVaccines": [
```

```
      {
        "id": "vac-01",
        "name": "Influenza (Flu)",
        "date": "Oct 20, 2026",
        "doseType": "Annual Shot",
        "icon": "syringe",
        "colorHex": "#FFC107"
      },
      {
        "id": "vac-02",
        "name": "Hepatitis B",
        "date": "Dec 05, 2026",
        "doseType": "Booster Dose",
        "icon": "shieldCheck",
        "colorHex": "#2196F3"
      }
    ],
    "completedVaccines": [
      {
        "id": "vac-03",
        "name": "COVID-19",
        "date": "Mar 12, 2024",
        "doseType": "Completed (2 Doses + Booster)",
        "icon": "checkCircle",
        "colorHex": "#4CAF50",
        "certificateUrl": "https://api.medirx.com/vaccines/cert-covid.pdf"
      }
    ]
  }
}
```

🧮 Module 7: Pediatric Dose Calculator

Calculates safe pediatric drug doses based on body weight, age, and medication concentration.

7.1 Calculate Safe Dose

- **Endpoint**: `POST /pediatric-dose/calculate`

Request Body


```
{
  "childWeightKg": 12.5,
  "childAgeMonths": 24,
  "medicineName": "Paracetamol",
  "formulation": "Syrup 120mg/5ml"
}
```


Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Dosage calculated according to WHO pediatric guidelines",
  "data": {
    "medicine": "Paracetamol",
    "calculatedDose": "2.5 ml",
    "frequency": "Every 6 hours",
```

```
    "maxDailyDose": "10 ml (in 24 hours)",
    "instructions": "Administer with oral syringe. Do not exceed 4 doses per day.",
    "warning": "Consult a pediatrician if fever persists after 48 hours."
  }
}
```

7.2 Get Recent Dosage Calculations

- **Endpoint**: `GET /pediatric-dose/recent`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Recent calculations retrieved",
  "data": [
    {
      "id": "calc-01",
      "medicine": "Paracetamol",
      "dose": "2.5 ml",
      "frequency": "Every 6 hours",
      "timestamp": "2026-10-12T14:20:00Z"
    },
    {
      "id": "calc-02",
      "medicine": "Amoxicillin Syrup",
      "dose": "4.0 ml",
      "frequency": "Every 8 hours",
      "timestamp": "2026-10-05T09:15:00Z"
    }
  ]
}
```

🤰 Module 8: Pregnancy Tracking

Provides gestational milestone computation, baby fruit size analogy, daily symptom logging, and trimester tips.

8.1 Get Pregnancy Status & Weekly Highlight

- **Endpoint**: `GET /pregnancy/status`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Pregnancy details retrieved",
  "data": {
    "currentWeek": 24,
    "currentDay": 3,
    "daysToGo": 112,
    "progressPercent": 60.0,
    "babySizeComparison": {
      "fruitName": "Cantaloupe",
      "approxLengthCm": 30.0,
      "approxWeightGrams": 600,
```

```
      "description": "Your baby is developing rapid eye movement (REM) sleep."
    },
    "weeklyTips": [
      {
        "id": "tip-01",
        "title": "Stay Hydrated",
        "desc": "Drinking plenty of water helps maintain amniotic fluid levels.",
        "icon": "droplets",
        "colorHex": "#03A9F4"
      },
      {
        "id": "tip-02",
        "title": "Gentle Exercise",
        "desc": "Try prenatal yoga or walking to stay active and reduce back pain.",
        "icon": "accessibility",
        "colorHex": "#4CAF50"
      }
    ]
  }
}
```

8.2 Log Pregnancy Symptoms & Mood

- **Endpoint**: `POST /pregnancy/symptoms`

Request Body


```
{
  "date": "2026-10-13",
  "symptoms": ["Nausea", "Back Pain", "Cravings"],
  "mood": "Calm",
  "fetalKickCount": 12,
  "waterIntakeLiters": 2.5
}
```


Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Daily symptoms logged successfully",
  "data": {
    "logId": "preg-log-1928",
    "savedAt": "2026-10-13T07:19:18Z"
  }
}
```

🌸 Module 9: Menstrual Cycle & Fertility

Calculates menstrual cycle phase, fertile window, ovulation forecasts, and flow symptoms.

9.1 Get Cycle Status & Predictions

- **Endpoint**: `GET /menstrual-cycle/status`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Menstrual cycle status calculated",
  "data": {
    "currentCycleDay": 14,
    "phase": "Ovulation Phase",
    "phaseDescription": "Ovulation phase - High chance of pregnancy",
    "cycleProgress": 0.5,
    "calendarMonth": "February 2026",
    "predictions": {
      "nextPeriod": {
        "title": "Next Period",
        "dateFormatted": "March 12, 2026",
        "daysRemaining": 14
      },
      "fertilityWindow": {
        "title": "Fertility Window",
        "rangeFormatted": "Feb 24 - Mar 01",
        "status": "High"
      }
    }
  }
}
```

9.2 Log Daily Cycle Symptoms & Flow

- **Endpoint**: `POST /menstrual-cycle/log`

Request Body


```
{
  "date": "2026-10-13",
  "flow": "Medium",
  "cramps": "Mild",
  "mood": "Happy",
  "waterGlasses": 8
}
```


Response `201 Created`


```
{
  "success": true,
  "statusCode": 201,
  "message": "Cycle data logged successfully",
  "data": {
    "entryDate": "2026-10-13",
    "recordedMetrics": ["Flow: Medium", "Cramps: Mild", "Mood: Happy", "Water: 8 glasses"]
  }
}
```

📍 Module 10: Nearest Medical Facilities

Performs geolocation-based radius search for hospitals, clinics, pharmacies, and emergency centers.

10.1 Search Nearby Facilities

- **Endpoint**: `GET /nearest-medical/facilities`
- **Query Parameters**: `lat=23.8103&lng=90.4125&radius=5&query=hospital`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Medical facilities retrieved",
  "data": [
    {
      "id": "fac-01",
      "name": "City General Hospital",
      "type": "hospital",
      "distance": "0.8 km away",
      "status": "Open 24/7",
      "phone": "+880 2-9834567",
      "latitude": 23.8150,
      "longitude": 90.4170,
      "icon": "building2",
      "colorHex": "#2196F3",
      "hasEmergency": true
    },
    {
      "id": "fac-02",
      "name": "MediCare Clinic",
      "type": "clinic",
      "distance": "1.2 km away",
      "status": "Closes at 10 PM",
      "phone": "+880 2-9876543",
      "latitude": 23.8190,
      "longitude": 90.4080,
      "icon": "home",
      "colorHex": "#4CAF50",
      "hasEmergency": false
    },
    {
      "id": "fac-03",
      "name": "Emergency Care Center",
      "type": "emergency",
      "distance": "2.5 km away",
      "status": "Open 24/7",
      "phone": "+880 9612-999999",
      "latitude": 23.8240,
      "longitude": 90.4200,
      "icon": "thermometer",
      "colorHex": "#00BCD4",
      "hasEmergency": true
    }
  ]
}
```

⭐ Module 11: Doctor Ratings & Reviews

Manages doctor ratings, feedback submission, and verified reviews.

11.1 Get Top Rated Doctors

- **Endpoint**: `GET /doctor-ratings`
- **Query Parameters**: `specialty=all&page=1&limit=10`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Doctor ratings loaded",
  "data": [
    {
      "doctorId": "doc-01",
      "name": "Dr. Sarah Johnson",
      "specialty": "Cardiologist",
      "rating": 4.8,
      "reviewsCount": 124,
      "avatarUrl": "https://i.pravatar.cc/150?u=sarah"
    },
    {
      "doctorId": "doc-02",
      "name": "Dr. Michael Chen",
      "specialty": "Pediatrician",
      "rating": 4.9,
      "reviewsCount": 89,
      "avatarUrl": "https://i.pravatar.cc/150?u=michael"
    },
    {
      "doctorId": "doc-03",
      "name": "Dr. Emily Davis",
      "specialty": "Dermatologist",
      "rating": 4.7,
      "reviewsCount": 156,
      "avatarUrl": "https://i.pravatar.cc/150?u=emily"
    }
  ]
}
```

11.2 Submit Doctor Review

- **Endpoint**: `POST /doctor-ratings/:doctorId/reviews`

Request Body


```
{
  "rating": 5.0,
  "review": "Very attentive and explained the prescription thoroughly. Highly recommended!",
  "appointmentId": "apt-89412"
}
```


Response `201 Created`


```
{
  "success": true,
  "statusCode": 201,
  "message": "Review submitted successfully",
  "data": {
    "reviewId": "rev-39120",
```

```
    "doctorId": "doc-01",
    "rating": 5.0,
    "createdAt": "2026-10-13T07:19:18Z"
  }
}
```

🤖 Module 12: AI Doctor Assistant

Provides interactive symptom analysis and healthcare conversation via an AI agent.

12.1 Send Message to AI Assistant

- **Endpoint**: `POST /ai-assistant/chat`

Request Body


```
{
  "sessionId": "chat-sess-908123",
  "message": "I have been feeling a bit dizzy lately."
}
```


Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "AI response generated",
  "data": {
    "sessionId": "chat-sess-908123",
    "reply": "I see. Dizziness can be caused by many factors such as dehydration, low blood pressure, or
inner ear issues. Have you been drinking enough water today?",
    "suggestedActions": [
      "Check Blood Pressure",
      "Drink 500ml Water",
      "Consult Cardiologist"
    ],
    "disclaimer": "This guidance is AI-generated and does not substitute a licensed physician consultation."
  }
}
```

12.2 Get AI Chat Session History

- **Endpoint**: `GET /ai-assistant/history/:sessionId`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Chat history fetched",
  "data": [
    {
      "sender": "ai",
      "message": "Hello! I am your AI Health Assistant. How can I help you today?",
      "timestamp": "2026-10-13T07:15:00Z"
```

```
    },
    {
      "sender": "user",
      "message": "I have been feeling a bit dizzy lately.",
      "timestamp": "2026-10-13T07:16:30Z"
    },
    {
      "sender": "ai",
      "message": "I see. Dizziness can be caused by many factors such as dehydration, low blood pressure, or
inner ear issues. Have you been drinking enough water today?",
      "timestamp": "2026-10-13T07:16:32Z"
    }
  ]
}
```

🔔 Module 13: Notification Center

Delivers appointment confirmations, medicine alarms, lab reports, and doctor alerts.

13.1 Get All Notifications

- **Endpoint**: `GET /notifications`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Notifications loaded",
  "data": [
    {
      "id": "notif-01",
      "title": "Appointment Confirmed",
      "message": "Your appointment with Dr. Sarah for tomorrow at 10:00 AM has been confirmed.",
      "time": "2 mins ago",
      "icon": "calendarCheck",
      "colorHex": "#4CAF50",
      "isRead": false,
      "actionRoute": "/scheduling/appointments/apt-89412"
    },
    {
      "id": "notif-02",
      "title": "Medication Reminder",
      "message": "It's time to take your Atorvastatin (10mg).",
      "time": "1 hour ago",
      "icon": "pill",
      "colorHex": "#009688",
      "isRead": false,
      "actionRoute": "/medications/active"
    },
    {
      "id": "notif-03",
      "title": "Lab Results Ready",
      "message": "Your recent blood test results are now available to view.",
      "time": "5 hours ago",
      "icon": "fileText",
      "colorHex": "#2196F3",
      "isRead": true,
      "actionRoute": "/profile/health-records/rec-02"
    },
```

```
    {
      "id": "notif-04",
      "title": "Prescription Renewed",
      "message": "Dr. Sarah has renewed your prescription for Metformin.",
      "time": "Yesterday",
      "icon": "checkCircle2",
      "colorHex": "#FF9800",
      "isRead": true,
      "actionRoute": "/medications/prescriptions/rx-001"
    },
    {
      "id": "notif-05",
      "title": "System Update",
      "message": "MediRx app has been updated to version 2.0 with new features.",
      "time": "3 days ago",
      "icon": "info",
      "colorHex": "#9E9E9E",
      "isRead": true,
      "actionRoute": null
    }
  ]
}
```

13.2 Mark Notification as Read

- **Endpoint**: `PATCH /notifications/:id/read`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Notification marked as read"
}
```

🗂️ Module 14: Patient Profile & Health Records

Manages personal demographic information, medical parameters, document storage, and security credentials.

14.1 Get Personal Information

- **Endpoint**: `GET /profile/personal-info`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Personal information retrieved",
  "data": {
    "name": "Druvo",
    "dob": "12 Dec 1990",
    "gender": "Male",
    "bloodGroup": "O+",
    "height": "175",
    "weight": "75",
    "email": "patient@example.com",
```

```
    "phone": "+1 234 567 890",
    "address": "123 Main St, Springfield",
    "avatarUrl": "https://i.pravatar.cc/150?u=patient"
  }
}
```

14.2 Update Personal Information

- **Endpoint**: `PUT /profile/personal-info`

Request Body

```
{
  "name": "Druvo",
  "dob": "12 Dec 1990",
  "gender": "Male",
  "bloodGroup": "O+",
  "height": "175",
  "weight": "75",
  "email": "patient@example.com",
  "phone": "+1 234 567 890",
  "address": "123 Main St, Springfield"
}
```

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Profile updated successfully",
  "data": {
    "name": "Druvo",
    "updatedAt": "2026-10-13T07:19:18Z"
  }
}
```

14.3 Upload Profile Avatar

- **Endpoint**: `POST /profile/avatar`
- **Content-Type**: `multipart/form-data`

Response `200 OK`

```
{
  "success": true,
  "statusCode": 200,
  "message": "Avatar uploaded successfully",
  "data": {
    "avatarUrl": "https://cdn.medirx.com/avatars/pat-001.png"
  }
}
```

14.4 Get Health Records

- **Endpoint**: `GET /profile/health-records`

Response `200 OK`


```
{
  "success": true,
  "statusCode": 200,
  "message": "Health records loaded",
  "data": [
    {
      "id": "rec-01",
      "type": "Prescription",
      "doctor": "Dr. Smith",
      "date": "10 Feb 2026",
      "title": "Viral Fever",
      "icon": "fileText",
      "colorHex": "#009688",
      "documentUrl": "https://api.medirx.com/records/rec-01.pdf"
    },
    {
      "id": "rec-02",
      "type": "Lab Report",
      "doctor": "City Lab",
      "date": "05 Feb 2026",
      "title": "Blood Test (CBC)",
      "icon": "flaskConical",
      "colorHex": "#FF9800",
      "documentUrl": "https://api.medirx.com/records/rec-02.pdf"
    },
    {
      "id": "rec-03",
      "type": "Prescription",
      "doctor": "Dr. Emily",
      "date": "20 Jan 2026",
      "title": "General Checkup",
      "icon": "fileText",
      "colorHex": "#009688",
      "documentUrl": "https://api.medirx.com/records/rec-03.pdf"
    },
    {
      "id": "rec-04",
      "type": "Vaccination",
      "doctor": "City Hospital",
      "date": "15 Dec 2025",
      "title": "Flu Shot",
      "icon": "syringe",
      "colorHex": "#9C27B0",
      "documentUrl": "https://api.medirx.com/records/rec-04.pdf"
    }
  ]
}
```

14.5 Change Password

- **Endpoint**: `POST /profile/change-password`

Request Body

```
{
  "currentPassword": "OldPassword123!",
  "newPassword": "NewSecurePassword456!"
}
```

Response 200 OK

```
{
  "success": true,
  "statusCode": 200,
  "message": "Password changed successfully"
}
```